

NEXOSxLPIN

1.4.0 Enterprise Hub · Platform Blueprint

Professional rebuild specification with UI/UX visual examples.

Designed so an operator or LLM coding agent can ingest this PDF and reconstruct the product.

Field	Value
Product root	C:\NEXOSxLPIN (local disk)
Stack	React 19 · TypeScript · Vite · Tailwind · Zustand · Leaflet · R3F
SME lenses	252 (specialized LENS_RULES each)
Congressional desks	56 full-depth training desks
Evidence model	Tri-state +1 / 0 / -1 · Layer-0 · Working Document
UI shells	Web multi-pane · Mobile single-column (persisted toggle)
Install	Node LTS → INSTALL.bat → START.bat → http://localhost:5173
Share zip	releases\NEXOSxLPIN-1.4.0-*.zip (no node_modules)

America First · Truth-Seeking · Evidence-first. Training desks are not legal advice. No secrets or client PII in sample packs.

1. Mission & non-negotiable principles

NEXOSxLPIN is a **Lily Pad Intelligence Network** workbench for story-driven investigation: map places, score claims, run Subject Matter Expert lenses, apply jurisdictional story rules, and publish only after Layer-0 integrity gates pass.

Hard rules

- Tri-state evidence only: **+1 Supported**, **0 Not proven**, **-1 Disputed**.
- Layer-0 pre-filter + one-shot ACK before high-stakes export / L3+ ladder.
- Working document is durable memory (decisions, notes, exports).
- Confirm before Apply SME scores to the evidence ledger (two-click).
- Explicit user-triggered export only — never auto-download.
- Prefer primary public records over social; no invented statutes or vote tallies.
- Product path on local disk (C:\), not OneDrive; npm.cmd on Windows.

Evidence language

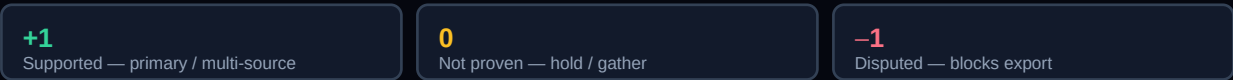


Figure 1 — Tri-state score badges used across Research, SME, Export, Status.

2. UI / UX visual system

Shell: night field #05070f, surfaces #121a2b, accent cyan #22d3ee, slate body text. Density: Dense / Compact / Roomy. Modes: Tabs · Tiles · Immersive (desktop). Header toggle forces **Web** or **Mobile** chrome.

2.1 Web shell (multi-pane)

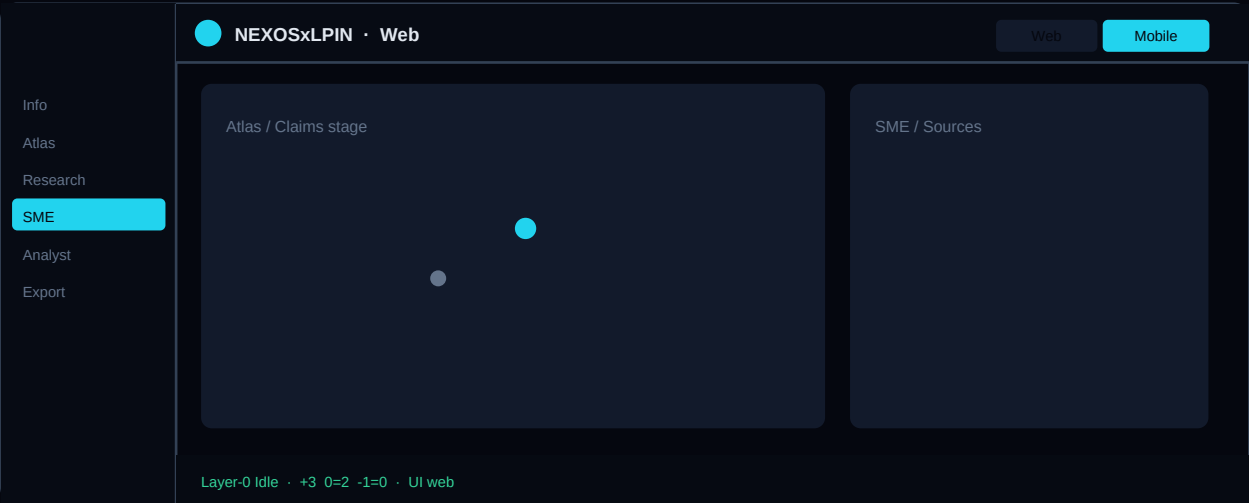


Figure 2 — Web layout: left module rail, header with brand + Web/Mobile + density, primary stage (Atlas/Claims), secondary SME/Sources, Layer-0 status bar.

2.2 Mobile shell (single-column)

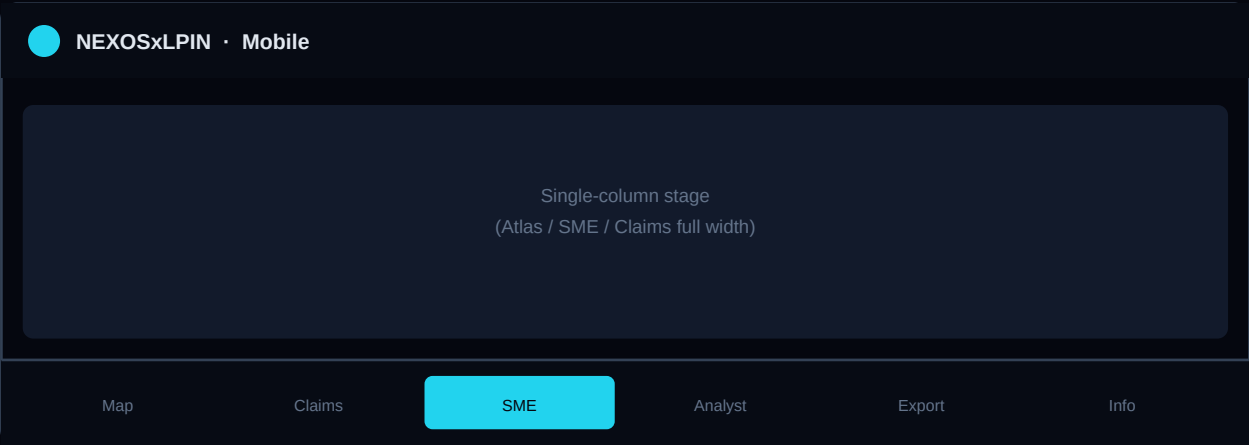


Figure 3 — Mobile: full-width stage, ≥44px bottom nav (Map · Claims · SME · Analyst · Export · Info). Music dock and dense chrome hidden. Immersive forced off.

2.3 Operator loop (UX journey)

Step	Surface	User action	Success signal
1	Picker / Switcher	Choose story or cong desk	useCasePicked · map pin active
2	Atlas	Review pins; click grey pin to switch	Active pin cyan; others grey
3	Research Hub	Score claims; open sources	Ledger +1/0/-1 · source links
4	SME Lenses	Select experts · Run selected	Batch briefings · posture badges
5	Design Lab	Set story rules / conditions	WD condition entry
6	Audit Ladder	Populate L0→L4	Ladder scores filled

Step	Surface	User action	Success signal
7	Export Kit	ACK Layer-0 · clear -1 · Download	File download · WD export log

3. Architecture & tech stack

Single-page app: Vite bundles React 19 + TypeScript. State: Zustand with persist (key nexos-lpin-v1). Spatial: Leaflet. 3D: React Three Fiber. Styling: Tailwind v4. Lint: oxlint. Test: vitest. Package manager: npm.cmd on Windows.

Directory map (product root C:\NEXOSxLPIN)

```
src/App.tsx – shell Web|Mobile
src/store/platformStore.ts – all runtime state + SME + uiMode
src/types/{core,sme,useCase}.ts – contracts
src/data/sme/ – 252 lenses (core + expansion packs)
src/lib/sme/{analyze,rules}.ts – engine + 252 specialized rules
src/data/useCases/ – catalog, stories, sims, congress packs
src/components/modules/* – 10 product surfaces
src/components/layout/* – Workspace, Sidebar, StatusBar
public/ – brand-mark.svg, favicon, icons
scripts/ – smoke, zip, brand, generators
docs/ – INSTALL, handoff, this blueprint
START.bat · INSTALL.bat · QUICKSTART.txt
```

Core state (platformStore)

Slice	Role
activeModule / workspace	Tabs·Tiles·Immersive, panes, fractions, locks
uiMode web mobile	Shell chrome; persisted
activeUseCaseId + setUseCase	Loads sim, evidence, sources, design, assets
evidence / researchNotes	Tri-state ledger + notes
layer0 / layer0AckToken	Integrity gate + one-shot ACK
workingDocument	Durable decisions / exports / SME briefs
activeSmeLensId / selectedSmeLensIds	Detail + multi-select
lastSmeBriefing(Set)	Advisor outputs
conditions / ladder / assets	Design Lab, Audit L0–4, Forge meshes

4. Module surface catalog (all working)

Module id	Label	Capability (1.4.0)
information	Information	Story body + SME help + full product guide
atlas	Atlas	All investigation pins; grey inactive; click switch; scene graph
design-lab	Design Lab	Story/jurisdiction condition matrices; apply → WD
research-hub	Research Hub	Claims, sources, notes; filter + bulk rescore
analyst	Analyst	Command runtime (sme/desk/export/ui)
sme-lenses	SME Lenses	252 experts; accordion; multi-run; confirm apply
audit-ladder	Audit Ladder	L0–L4 progression with scores/notes
procedural-forge	Procedural Forge	Story-linked mesh gen; Unity C# + Three export
massing-viewer	Massing Viewer	Live R3F runtime for active asset
export-kit	Export Kit	Preflight checklist; ACK; explicit download

SME Lenses UX (must preserve)

- Domain accordion with aria-expanded / aria-controls
- Checkbox + label association; click tagline opens detail
- Sticky filter + Expand/Collapse; horizontal selected chips + Clear all
- Recommended chips from tag relevance
- Run SME analysis · Run selected · Commit WD · Apply scores (confirm)
- Batch briefings list with short name, urgency, posture badge

5. SME system — 252 lenses

Each lens is a professional research persona. Engine: deterministic offline analyzeWithLens + specialized LENS_RULES[id]. assertAllLensesHaveRules() must return [].

Domain	Count	Focus
core-governance	20	Evidence, Layer-0, narrative, custody, harm
public-records	20	FOIA, dockets, permits, title, bid tabs
jurisdiction	20	Authority maps, preemption, admin law
oversight	20	Influence, ethics, IG, lobby, grants
sector-regulatory	22	Env, health, cyber, export, finance...
method-process	11	Verify, sources, metrics, uncertainty
mechanical-engineering	17	Statics, fatigue, CFD, NVH...
civil-structural	11	Structural, seismic, bridge, flood
electrical-electronics	14	Power, EMC, embedded, battery
chemical-process	8	Process, LOPA, catalysis
aerospace-defense-tech	8	Flight, GNC, mission assurance
materials-manufacturing	11	Metallurgy, AM, welding, quality
energy-nuclear	8	Grid, nuclear safety, H2, CCS
biomedical-systems	8	Devices, clinical, biocompatibility
computing-cyberphysical	14	CPS, security, ML systems, SBOM
mathematics-statistics	20	Stats, causal, Bayesian, PDE...
theoretical-physics	14	Classical→QFT literacy, regimes
applied-physical-sciences	6	Metrology, remote sensing, acoustics
TOTAL	252	

Lens field contract (every NEW lens)

id, short (≤ 18), name, domain, tagline, description (≥ 2 sentences), persona (title, credential, voice, ≥ 3 principles), focusTags ≥ 6 , questionBank ≥ 3 , preferredSources ≥ 2 , publishGates ≥ 1 , highStakes bool, LENS_RULES[id] specialized.

6. Congressional / industry desks — 56

Family congressional. Each desk: UseCaseReport (≥5 mixed claims), InvestigationStory, ≥5–6 official ActiveSources, simulation + mapPin (jittered). Framing: training / industry-effect — not legal advice.

cong-01...20 original · cong-21...40 1.3 expansion · cong-41...56 1.4 themes (AI chip export, biometric procurement, COOP, OSS federal, med-device cyber, AWS dual-use, carbon offsets, digital ID, freight data, hardrock, broadcast, 340B, BVLOS, SMR, tribal energy, debris).

Source hierarchy

- 1. Congress.gov bill/search
- 2. Named agency primary (FDA, BIS, FCC, FEMA...)
- 3. GAO
- 4. CRS
- 5. GovInfo / authenticated pubs

7. Analyst command surface

```
help · modules · open <module> · ack [reason] · layer0 <action|status>
score <+1|0|-1> <title> :: <summary>
conditions apply · ladder <0-4> · forge generate <type>
sme list|tech|count|domains|search <q>|select <id>|run|run-domain <d>|<lens-id>
desk list|cong · evidence summary · export check · ui web|mobile · status
```

8. Key TypeScript contracts (rebuild skeleton)

```
type EvidenceScore = 1 | 0 | -1
type ModuleId = 'information' | 'atlas' | 'design-lab' | 'research-hub' | 'analyst' | 'sme-lenses' | 'audit-ladder' | 'procedural-fo
rge' | 'massing-viewer' | 'export-kit'
interface SmeLens { id; short; name; domain; tagline; description; persona; focusTags; questionBank; preferredSources;
publishGates; highStakes }
interface SmeBriefing { posture; urgency; claimReads; actions; workingDocMarkdown; stats }
interface UseCaseProfile { id; label; family; report?; mapPin?; defaultOpen; paneWeights... }
WorkspaceViewMode = 'tabs' | 'tiles' | 'immersive' · uiMode = 'web' | 'mobile'
```

9. Install & verify

```
cd /d C:\NEXOSxLPIN
INSTALL.bat :: npm.cmd install && npm.cmd run build
START.bat :: npm.cmd run dev → http://localhost:5173
npm.cmd run test
npm.cmd run lint
npm.cmd run build
node scripts\smoke-sme-congress.mjs
powershell -File scripts\create-desktop-shortcut.ps1
```

Acceptance gates (1.4.0)

Gate	Expect
SME_LENSES.length	252
assertAllLensesHaveRules()	[]
congressionalDesks().length	56
Per-domain counts	Exact table in §5
Unique ids/shorts	No collisions
Export	Blocked on -1; needs ACK
Apply scores	Two-click confirm
Bundle	Chunked (main ■ 1.8MB)

10. LLM rebuild prompt (copy into coding agent)

Paste the following into Claude / Grok / Cursor / Codex with this PDF attached:

You are building NEXOSxLPIN 1.4.0 Enterprise Hub from the attached Platform Blueprint PDF. Stack: React 19 + TypeScript + Vite + Tailwind + Zustand (persist) + Leaflet + R3F. Implement all 10 modules, Web|Mobile uiMode, tri-state evidence +1/0/-1, Layer-0 ACK, working document, SME engine with 252 specialized lenses/rules (domain counts exact), 56 congressional training desks with reports/stories/sources/sims, Analyst commands listed in §7, Export Kit preflight, confirm-before-apply scores, chunked Vite manualChunks, INSTALL.bat/START.bat, smoke tests asserting 252/56/rules[]. cwd local disk only; npm.cmd on Windows; no secrets/PII; training desks not legal advice; no invented statutes. Prefer expansion packs over rewriting core ids. Match UI wireframes in §2 (night theme, cyan accent, status bar Layer-0).

11. Benchmark snapshot (post-audit 1.4.0)

Metric	Value
SME lenses / rules	252 / 252
Congressional desks	56
Vitest	16/16 (analyze + smoke)
Main JS (approx)	~294 kB entry · chunked vendors
Cold install	Node LTS + INSTALL + START
Desktop	NEXOSxLPIN.lnk → START.bat · v140.ico

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